

⇒ Find $P(x=3, y=-3)$

$$P(x=3, y=-3) = f(3, -3) =$$

New chap 105 ~~continuous~~

1) Discrete Prob distribution

It consists following PD

1. Binomial PD
2. Multinomial PD
3. Poisson PD
4. Approximation Binomial PD
5. Geometric PD
6. Negative PD
7. Hypogeometric PD

2.) **BINOMIAL PROB DISTRIBUTION :**

Type ⇒ Discrete

Notation ⇒ $bn(x:np)$

parameter ⇒ n, p

$$PMF = f(x) = {}^n C_x p^x q^{n-x} ; x=0, 1, 2, \dots$$

$$MEAN \quad E(\bar{X}) = np$$

$$S.D = \sigma = \sqrt{npq}$$

where n is the total no of independent trials,

p is the prob of success

q is the prob for failure

use when there are only 2 ops

probable to success 'p'
opposite will be failure

$$\sum_{i=0}^{\infty} f(x) = 1$$
$$f(x) \geq 0$$
$$p + q = 1$$
$$q = 1 - p$$
$$p = 1 - q$$

Q) The probability that a patient survive from disease is 40%. If 15 patients are registered from this disease. what is a probability that exactly 5 patient will survive.

$$n = 15, \quad p = 40\% = 0.4, \quad q = 0.6$$

i) $p(x=5) = ?$

$$f(x=5) = \binom{n}{x} p^x q^{n-x}$$
$$\Rightarrow \binom{15}{5} (0.4)^5 (0.6)^{15-5}$$
$$\Rightarrow \binom{15}{5} (0.4)^5 (0.6)^{10}$$
$$\Rightarrow 0.1859$$

ii) $p(x=2) = ?$

$$f(x=2) = \binom{15}{2} (0.4)^2 (0.6)^{15-2}$$
$$= \binom{15}{2} (0.4)^2 (0.6)^{13}$$
$$\Rightarrow 0.021$$

Q) 12 students are given 2 identical speakers to listen to for differences if any suppose that most of the student answered by guessing technique find the probability that 3 students claimed that there are some difference b/w 2 speakers

$$n = 12$$

$$p = ? \text{ it will be } 50\% \text{ so } \Rightarrow p = 0.5$$

$$q = ? \text{ also } 50\% \Rightarrow q = 0.5$$

$$p(x=3) = ?$$

$$\begin{aligned} f(x=3) &= {}^n C_x p^x q^{n-x} \\ &= ({}^{12} C_3) (0.5)^3 (0.5)^{12-3} \\ &\Rightarrow 0.053 \end{aligned}$$

Q) A traffic control engineer report that 75% of the traffic passing through a check point are from within the state. What is the prob that fewer than 4 of the next 9 traffics are from out of the state

$$n = 9, \quad q = 0.75, \quad p = 0.25, \quad p(x=4)$$

we have to find p now bcz of out of the state

$$f(x \leq 4) \quad \text{for fewer some will find } x=0,1,2,3 \text{ values}$$

$$f(x=0) = ({}^9 C_0) (0.25)^0 (0.75)^{9-0}$$

=

$$f(x=1) \Rightarrow$$

$$f(x=2) \Rightarrow$$

$$f(x=3) \Rightarrow$$

PMF = Probability Mass Function

MULTINOMIAL PROB DISTRIBUTION:

Type $\Rightarrow$ discrete

Notation $\Rightarrow$ Multinomial $(x_1!, x_2! \dots p_1, p_2 \dots n)$

Parameter $\Rightarrow x, p$

$$\text{PMF} = f(x) = f(x_1! x_2! \dots p_1 p_2 \dots n) = \frac{n!}{x_1! x_2! \dots} p_1^{x_1} p_2^{x_2} \dots$$

More than 2 option we will use multinomial distribution

Q) For a certain airport containing 3 runways the prob that individual runways are as follows

$$\text{Runway \#1} = p_1 = 2/9$$

$$\text{Runway \#2} = p_2 = 1/9$$

$$\text{Runway \#3} = p_3 = 1/18$$

What is the prob that 6 randomly arriving flight are distributed in the following fashion

$$\text{Runway \#1} = 2, \quad \text{Runway \#2} = 1$$

$$\text{Runway \#3} = 3$$

$$n = 6$$

$$x_1 = 2 = p_1 = 2/9$$

$$x_2 = 1 = p_2 = 1/9$$

$$x_3 = 3 = p_3 = 1/18$$

$$M \sim (x_1! x_2! \dots p_1 p_2 \dots n)$$

$$M \sim (2, 1, 3, 2/9, 1/9, 1/18, 6)$$

$$= \frac{n!}{x_1! x_2! x_3!} p_1^{x_1} p_2^{x_2} p_3^{x_3}$$

$$x_1! x_2! x_3!$$

$$\Rightarrow \frac{6!}{2!1!3!} \left[(2/9)^2 (1/6)^1 (11/18)^3 \right]$$

$$\Rightarrow 0.118$$

Q) A student drive to university he encounters a traffic signal. This traffic signal still of green 35 sec still for yellow for 5 sec & still for red for 1 min. let x_1 is the no of times for yellow light and x_2 is the no of time of green and x_3 is the no of time for red light. Find joint prob dist formula for x_1, x_2, x_3

$$G = 35 \text{ sec} \Rightarrow P(G) = 35/100 = 0.35$$

$$Y = 5 \text{ sec} \Rightarrow P(Y) = 5/100 = 0.05$$

$$R = \frac{60 \text{ sec} = 1 \text{ min}}{100} \Rightarrow P(R) = 60/100 = 0.6$$

Mn

$$= \frac{n!}{x_1! x_2! x_3!} P_1^{x_1} P_2^{x_2} P_3^{x_3}$$

$$x_1 = x_1$$

$$x_2 = x_2$$

$$x_3 = x_3$$

$$\Rightarrow \frac{100!}{x_1! x_2! x_3!} (0.35)^{x_1} (0.05)^{x_2} (0.6)^{x_3}$$

General formula created because

x_1, x_2, x_3 values are not given.

Q) A/c 16 US surveys of 4 million workers, 5.8% testing +ve for drugs of those testing +ve 22.5% were cocaine users and ~~22.5%~~ 54.4% are marijuana users

- 1.) what is the prob that of 10 workers testing +ve to our cocaine user 5 are marijuana user & rest of the worker use other drugs
- 2.) what is the prob that 10 workers testing +ve & all are marijuana users.
- 3.) what is the prob that 10 workers testing +ve non are cocaine users. $1 - 0.225$

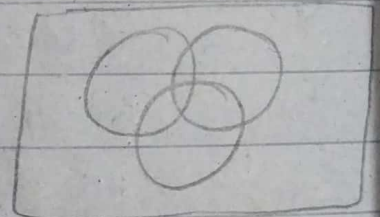
$$n = 4000000$$

$$D = 5.8\% = 0.058$$

$$C = 22.5\% = 0.225$$

$$M = 54.4\% = 0.544$$

Venn diagram



Q) The prob are 0.4, 0.2, 0.3 respectively. that a delegates
to a certain conference are arrived by ~~plane~~^{air}, by bus, by
car & by train. What is the prob that among 9 delegates
randomly selected at this conference. 3 arrive by air.
3 by bus, 1 by car & 2 by train.

$$\begin{array}{l} p_1 = 0.4 \\ p_2 = 0.2 \\ p_3 = 0.3 \\ p_4 = 0.1 \end{array} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} +$$

For multinomial λ_{mp}
pg# 175 S.20, S.21, S.22

For Binomial
pg# 174 S.6
S.6, S.8, S.9
S.11, S.15

POISSON PROB DISTRIBUTION :

Type $\Rightarrow$ Discrete

Notation $\Rightarrow \text{Poi} \sim (x : \lambda)$

λ or σ or μ

Parameter = λ

$$\text{PMF} \Rightarrow f(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} \quad \text{where } x = 0, 1, 2, \dots$$

$$E(\bar{x}) = \lambda, \quad \sigma^2 = E(v) = \lambda \text{ (mean)}$$

$$\text{S.D} = \sigma = \sqrt{\lambda}$$

$$\lambda t = \text{for timing}$$

$$\lambda = \lambda t$$

If Mean & Average word comes in any Q the poisson dis will be applied

Q) During a lab testing the average number of defective elements passing through a counter in 1 msec is 4 what is the prob that 2 particles enter in the counter in a given millisecond

$$\lambda = 4 \times 1 = 4$$

$$P(x=2) = ?$$

$$\begin{aligned} f(x) &= \frac{e^{-\lambda} \cdot \lambda^x}{x!} \\ &= \frac{e^{-4} \cdot 4^2}{2!} \end{aligned}$$

$$f(x) \Rightarrow 0.14$$

Now for 6 particles,

$$\begin{aligned} f(x) &= \frac{e^{-\lambda} \cdot \lambda^x}{x!} \\ &= \frac{e^{-4} \cdot 4^6}{6!} \end{aligned}$$

$$f(x) = 0.104$$

Q) A typist make ^{λ} 2 error ^{1 page} per page on the average. What is the prob that on the next page. He or she will make 1.) 0 error 2) 2 error

$$\lambda = 2 \text{ per page so } 2 \times 1 = 2$$

$$1) f(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} \Rightarrow \frac{e^{-2} \cdot \lambda^0}{0!} \Rightarrow \boxed{0.135}$$

$$2) f(x=2) \Rightarrow \frac{e^{-2} \cdot \lambda^2}{2!} = \boxed{0.270}$$

Q) In a manufacturing process where last product are produced ~~on the~~ It is known that on the average 1 in every 1000 of these items produced has 1 or more defective. What is the prob that at a random ^{sample} 8000 will produce fewer than 4 items are defective.

Approximation Case:

$$b \sim p$$

① $n \rightarrow \infty$ 100, 25, 1000

② $p \rightarrow \text{small}$ 0.0001, 0.001

($n \rightarrow \infty$ means
n value is large)

$$b \sim p$$

$$\boxed{np = 1}$$

Q The prob that a person will not survive from a certain type of viral infection is 0.002. Find the prob. that fewer than 5 of the next 2000 so infected will not survive.

$$n = 2000$$

$$p = 0.002$$

n is large & p is small so condition satisfies
 $P(\text{Fewer } 5)$

$$P(X < 5)$$

so we will use formula

$$np = 1$$

$$0.002 \times 2000 = 1$$

$$\boxed{\lambda = 1}$$

$$P(X < 5), \quad X = 0, 1, 2, 3, 4$$

$$P(X=0) \Rightarrow \frac{e^{-\lambda} \cdot \lambda^0}{0!} \Rightarrow$$

$$P(X=1) \Rightarrow \frac{e^{-\lambda} \cdot \lambda^1}{1!} \Rightarrow$$

$$P(X=2) \Rightarrow \frac{e^{-\lambda} \cdot \lambda^2}{2!} \Rightarrow$$

$$P(X=3) \Rightarrow \frac{e^{-\lambda} \cdot \lambda^3}{3!} \Rightarrow$$

$$P(X=4) = \frac{e^{-\lambda} \cdot \lambda^4}{4!}$$

$\Rightarrow$

$$\sum_{x=0}^4 f(x) = \text{add all the } x$$

Imp

Q In a certain industrial facilities accident occur infrequently. It is known that the Prob of an accident on any given day is 0.005. Suppose that accidents are independent to each other

1. Find the Prob that any given period of 400 days there will be an accident on one day
2. What is a Prob that there are almost 3 days with an accident

$$p = 0.005$$

$$n = 400$$

$$1. P(X=1)$$

$$2. P(X \leq \overset{\text{almost}}{3}) \text{ so } X = 0, 1, 2, 3$$

$$np = \lambda$$

$$400 \times 0.005 = \lambda$$

$$\lambda = 2$$

$$P(X=1) = \frac{e^{-\lambda} \cdot \lambda^1}{1!}$$

The Prob that a person will not survive certain type of infection is 0.001 of the next 4000 so infected. Find the prob ~~at least~~ exactly 5 patient- will survive. Find the mean, variance & standard deviation of the distribution.

$$n = 0.001$$

$$p = 4000$$

$$\boxed{\lambda = 4}$$

$$P(X=5) \quad x = 0, 1, 2, 3, 4, 5$$

we will use Geometric jahan time ka wait karna pata

It will always start from 1 not 0 (value of x)

Geometric Probability Distribution:

- (i) Type = discrete
- (ii) Notation $g \sim (x: p)$
- (iii) p
- (iv) PMF = $f(x) = pq^{x-1}$ $x = 1, 2, 3, 4, \dots$
- (v) $E(\bar{x}) = \text{mean} = 1/p$
- (vi) variance (σ^2) = $1-p/p^2$
- (vii) standard deviation = $\sigma = \frac{\sqrt{1-p}}{p}$

Q In a certain manufacturing process it is known that on the average 1 in every 100 items is defective. What is the prob. that the 5th item checked by quality control engineer is the first defective item found

$$p = \frac{1}{100} = 0.01$$

(100 items are defective)

$$q = 0.99 \quad \leftarrow 1 - 0.01 = 1 - p$$

pehle defective item 5 pay mila yani kay time calculate karke then hum 5 tak phunchay so wait hora hai be3 woh sab ko check karay hue jayega that why we are using geometric

$$P(X=5)$$

$$f(x=5) = (0.01)(0.99)^{5-1} \\ = 0.0096$$

amp
 Q The prob that a student pass prob of stat's exam is 0.7. Find the Prob that the student will pass this exam on 3rd trial

$$p = 0.7$$

$$q = 0.3$$

$$P(x=3) = (0.7)(0.3)^{3-1}$$

$$P(x=3) = 0.063$$

2. Before 4th trial so

$$P(x)$$

$$x = 1, 2, 3,$$

$$P(x=1) =$$

-

=

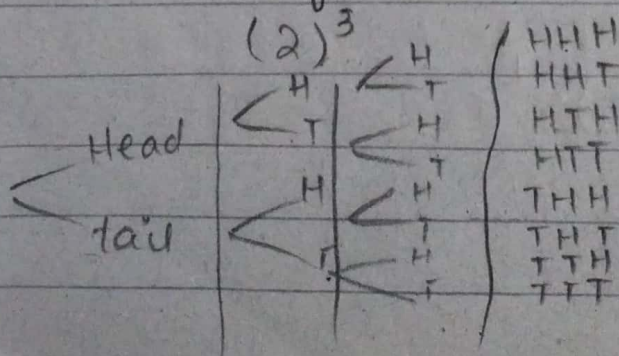
-

-

-

amp

Q Three student toss a fair coin & the odd man pay for lunch. If the coin all turn up the same they are toss again. Find the Prob that fewer than 4 toss are required



$$p = \frac{6}{8}$$

$$q = 1 -$$

$$P(\text{Fewer than 4}) \Rightarrow x = 1, 2, 3$$

Continuous RANDOM VARIABLE:

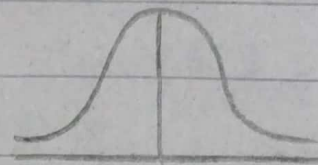
1st properly:

Normal Probability Distribution:

type: Continuous

$$N \sim (x: \mu, \sigma)$$

$$\text{PDF: } f(x: \mu, \sigma) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2} \quad -\infty \leq x \leq +\infty$$



$$-\infty \leq x \leq +\infty$$

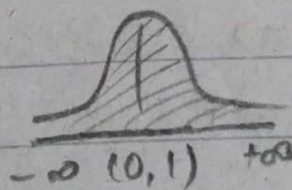
In Z distribution

mean = 0, standard deviation = 1.

$$N \sim \text{SND} \Rightarrow Z = \frac{x - \mu}{\sigma}$$

$$A_1 = 1$$

Area under the normal curve = 1
can be less than 1



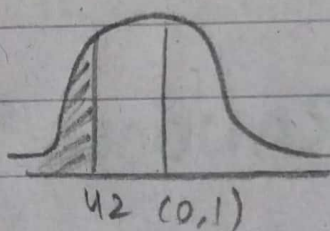
Q Given a normal Distribution with mean = 40 & SD = 6
Find the prob or Area under the normal curve that
assume a value Below 42

$$\mu = 40 \quad \sigma = 6$$

$$P(x < 42) = ?$$

$$z = \frac{x - \mu}{\sigma} \Rightarrow \frac{42 - 40}{6} \Rightarrow 0.33$$

$$z = 0.33 = 0.6293 \rightarrow \text{from the table}$$



Area under the
normal curve

ii) Find the Prob above 27

$$P(x > 27)$$

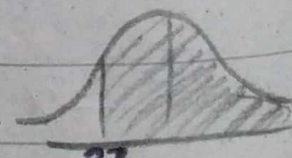
from $\frac{0.06}{-2.1}$

$$z = \frac{27 - 40}{6} = -2.16 \Rightarrow 0.0154$$

Now will find Prob

$$P(x > 27) \sim 1 - P(x < 27)$$

$$1 - 0.0154 \Rightarrow 0.9846$$



iii) Anywhere from 42 and 51

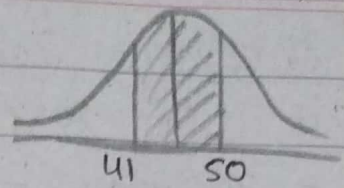
$$P(x > 42) \Rightarrow 0.33 \Rightarrow 0.6293 = P(x_1) = P(z_1)$$

$$P(x < 51) \Rightarrow 1.83 \Rightarrow 0.9664 = P(x_2) = P(z_2)$$

$$P(42 \leq x \leq 51)$$

$$P(z) = P(z_2) - P(z_1) \\ = 0.9664 - 0.6293$$

$$P(z) = 0.337$$



Q) The Burning time of an experimental rocket is a continuous random variable having normally distributed with mean 4.76 sec & SD = 0.04. What is the Prob of Area under the normal curve. This kind of rocket will burn-up more than 4.8 sec

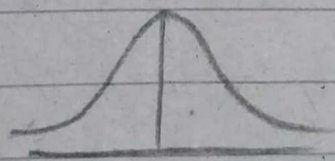
$$\mu = 4.76$$

$$\sigma = 0.04$$

$$P(x > 4.8)$$

$$z = \frac{4.8 - 4.76}{0.04} = 1 \Rightarrow 0.8413$$

$$P(x > 4.8) = 1 - 0.8413 \Rightarrow 0.1587$$

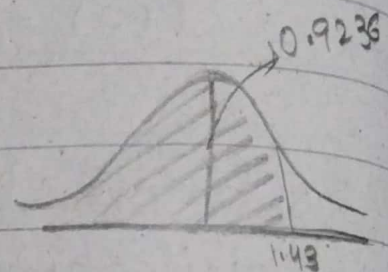


Q) Given a standard normal Distr. Find Area Under the Curve

1. To the left of $z = 1.43$

2. Between $z = -0.48$ and $z = 1.74$

1. $z = 1.43 \Rightarrow 0.9236$



2. $(z_1) = -0.48 = 0.3156$

$(z_2) = 1.74 = 0.9591$

$P(z) = z_2 - z_1$
=

Q) When Data is given

2, 3, 5, 7, 8, 10 $\rightarrow$ From the given data we will first find mean and SD

$P(x > 11)$

mean $\Rightarrow \bar{x} = \frac{\sum x_i}{n} \Rightarrow \frac{35}{6} \Rightarrow 5.83$

$SD = \sqrt{\left(\frac{\sum x^2}{n}\right) - \left(\frac{\sum x}{n}\right)^2}$

$\Rightarrow \frac{251}{6} - 33.98$

2	4
3	9
5	25
7	49
8	64
10	100

Q An Electrical firm manufacturing light bulb. have a life is normally distributed with mean 800hrs and variance is 1600hrs. Find the Prob that the bulb work b/w 778 and 834 hrs.

$$\mu = 800$$

$$SD = \sqrt{1600} = 40 \text{ hrs}$$

$$P(778 \leq X \leq 834)$$

$$P(X_1) = P(Z_1) = P(X > 778) = \frac{778 - 800}{40} = -0.55$$
$$\Rightarrow 0.2912$$

$$P(X_2) = P(Z_2) = P(X < 834) = \frac{834 - 800}{40} = 0.85$$
$$\Rightarrow 0.8023$$

$$P(Z) = Z_2 - Z_1$$
$$= 0.511$$

1/Jan/2025

practice Que

Normal distribution

Q) A soft drink machine is regulated so that it discharge an average of 200 millilitre per cups if the amount of the drink is normally distributed with standard deviation 15 ml per cup. e) what is the Prob that the cup will contain more than 224 millilitre

$$\mu = 200 \text{ ml}$$

$$\sigma = 15 \text{ ml}$$

$$P(X > 224) = ?$$

$$Z = \frac{224 - 200}{15} = 1.6$$

$$Z = 0.9452$$

$$\Rightarrow 1 - 0.9452$$

$$P(X > 224) = 0.0548$$

Normal $\rightarrow$ continuous

Binomial $\rightarrow$ discrete

APPROXIMATION CASE FOR NORMAL DISTRIBUTION

i) $n \rightarrow \infty$

n should be large

ii) $np \geq 10$

iii) $nq \geq 10$

These are three conditions or agar 2 bhi satisfy hojati hain toh $(N \sim b)$ normal mean will be equal to approx binomial mean

$$\mu = np$$
$$\sigma = \sqrt{npq}$$

correction factor $= \pm 0.5$

applies when discrete

continuous mein jaye

yea continuous $\rightarrow$ discrete

$$① P(x=a) \sim P(a-0.5 \leq x \leq a+0.5)$$

transform when this condition applies

$$② P(x < a) \sim P(x \leq a-0.5)$$

$$③ P(x > a) \sim P(x > a+0.5)$$

$$④ P(x \leq a) \sim P(x \leq a+0.5)$$

$$⑤ P(x \geq a) \sim P(x \geq a-0.5)$$

$$⑥ P(a \leq x \leq b) \sim P(a-0.5 \leq x \leq b+0.5)$$

Q) A fair coin toss 400 times. Find the Prob that less than 209 has been occur

$$n = 400$$

$$P(H) = 0.5$$

$$P(T) = 0.5$$

$$P(x < 209) = ?$$

$$⑦ np \geq 10$$

$$400 \times 0.5 \geq 10 \Rightarrow \boxed{200 \geq 10}$$

$$⑧ nq \geq 10$$

$$400 \times 0.5 \geq 10 \Rightarrow \boxed{200 \geq 10}$$

$$N \sim b$$

$$\mu = np$$

$$\boxed{\mu = 200}$$

$$\sigma = \sqrt{npq}$$

$$\boxed{\sigma = 10}$$

$$= \sqrt{400 \times 0.5 \times 0.5}$$

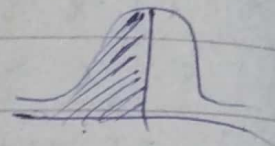
$$\Rightarrow \boxed{Z = \frac{x - np}{\sqrt{npq}}} \text{ formula}$$

$$P(X < 200) \sim P(X < 208.5)$$

$$P(X < 208.5) = Z = \frac{x_1 - np}{\sqrt{npq}}$$

$$= \frac{208.5 - 200}{10} = 0.85$$

$$P(X < 208.5) \Rightarrow 0.8023$$



ii) Exactly 205 has been occurred

$$P(X = 205) \sim P\left(\overset{x_1}{204.5} \leq X \leq \overset{x_2}{205.5}\right)$$

$$P(21) = \frac{204.5 - 200}{10} = 0.45 \Rightarrow 0.6736$$

$$P(22) = \frac{205.5 - 200}{10} = 0.55 \Rightarrow 0.7088$$

$$P \Rightarrow Z_2 - Z_1$$

$$\Rightarrow 0.0352$$

iii) $P(X \geq 200)$

$$P(X \geq 200) \sim P(X \geq 199.5)$$

$$P(X \geq 199.5) \Rightarrow Z = \frac{x_1 - np}{\sqrt{npq}}$$

$$= \frac{199.5 - 200}{10} = -0.05$$

$$\Rightarrow 0.4801$$

$$\Rightarrow 1 - 0.4801$$

$$\Rightarrow 0.5199$$

pg 217 Q 6.25, 6.26, 6.27, 6.28, 6.33, 6.34
For normal approximation

Q) A multiple choice Questioner has 200 Questions each Question with 4 possible answer of which only 1 is correct answer. Find the Prob from 25 - 30 correct answers for 80 of the 200 problems about which the student has no knowledge.

$$P(25 \leq x \leq 30) = ? \quad n = 200 \quad 80$$

$$p = 1/4$$

$$q = 3/4$$

all 3

$$80 \times 0.25 \geq 10$$

$$80 \times 0.75 \geq 10$$

} satisfy conditions

so $N \sim b$ (normal approximation in binomial)

$$\mu = np$$

$$\boxed{\mu = 20}$$

$$\sigma = \sqrt{npq}$$

$$\boxed{\sigma = 3.87}$$

$$\Rightarrow \sqrt{80 \times 0.25 \times 0.75}$$

$$P(25 \leq x \leq 30) \sim P(24.5 \leq x \leq 30.5)$$

$$P(x \geq 24.5) = z_1 = \frac{24.5 - \overset{20}{80}}{3.87} \Rightarrow 1.162 \Rightarrow 0.8770$$

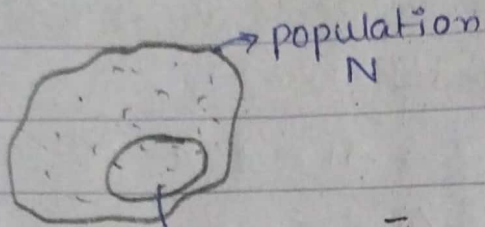
$$z_2 = \frac{30.5 - \overset{20}{80}}{3.87} \Rightarrow 2.71 \Rightarrow 0.9966$$

$$z_2 - z_1 = 0.9966 - 0.8770$$

$$\boxed{\Rightarrow 0.1196}$$

SAMPLING

The process of selecting a small part from large population such selected small part which show all the characteristic of large population.



$$N = \mu = \frac{\sum x_i}{N} \rightarrow \text{mean parameter (info from population)}$$

$$\bar{x} = \frac{\sum x_i}{n}$$

info from sample = statistics

① Sampling without Replacement

$$(S.W.O.R)^n C_r = \frac{n!}{(n-r)!r!}$$

② Sampling with Replacement

$$(S.W.R)^n C_r = (n)^r \text{ or } (n)^n$$

basic properties of SWOR

(i) $\mu = \bar{x}$

population mean always equal to sample mean

(ii) $V(\bar{x}) = \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1}$

sample variance

Where σ^2 = population variance

N = no of population

n = sample no

basic properties of SWR

(i) $\mu = \bar{x}$

(ii) $V(\bar{x}) = \frac{\sigma^2}{n}$

where σ^2 = population variance

if size 3 so (x, y, z)

Q) A population consists 5 number 0, 2, 4, 6, 8. Find the following quantities

- 1) list all possible samples of size 2 without replacement ^{sd(x,y)}
- 2) Find the mean of each sample
- 3) Construct sampling distribution of $(\bar{x})$
- 4) show that $\mu = \bar{x}$

$$N = 0, 2, 4, 6, 8$$

$$n = 5$$

$$n = r = 2$$

W.O.R

$${}^n C_r = \frac{n!}{r!(n-r)!}$$

$$\Rightarrow \frac{5!}{2!(5-2)!}$$

$$\Rightarrow \boxed{{}^5 C_2 = 10}$$

# Sample	Sample ^{size=2}	Sample $\bar{x}$
1	(0, 2)	$\frac{0+2}{2} = 1$
2	(0, 4)	2
3	(0, 6)	3
4	(0, 8)	4
5	(2, 4)	3
6	(2, 6)	4
7	(2, 8)	5
8	(4, 6)	5
9	(4, 8)	6
10	(6, 8)	7

Now constructing distribution

Sample	$\bar{x}$	frequency y	$x_i f_i$
1	1	1	1
2	2	1	2
3	3	2	6
4	4	2	8
5	5	2	10
6	6	1	6
7	7	1	7
		<u>10</u>	<u>40</u>

Σ frequency should be equal to n or

$$\bar{x} = \frac{\sum x_i f_i}{\sum f_i} = \frac{40}{10} = \boxed{4}$$

$$M = \frac{\sum x_i}{N} = \frac{0+2+4+6+8}{5} \Rightarrow \frac{20}{5} = \boxed{4}$$

$$M = \bar{x}$$

$$\boxed{4 = 4}$$

Hence Proved!

Q) Now find variance of S.W.O.R

$$V(\bar{x}) = \frac{s^2}{n} \cdot \frac{N-n}{N-1}$$

$$s^2 = \frac{\sum x_i^2}{N} - \left(\frac{\sum x_i}{N} \right)^2$$

$$\Rightarrow \frac{120}{5} - \left(\frac{20}{5} \right)^2 \Rightarrow 8$$

(N) bcz hum
population say
nikal rahay hain

with replacement = repetition allowed
 without = not allowed

$$V(\bar{x}) = \frac{8}{2} \cdot \frac{5-2}{5-1} \Rightarrow 3$$

Q consider

2, 4, 6, 3, 8 using sampling with replacement
 find the.

1. list all possible samples of size 2
2. calculate mean of each sample
3. construct sampling distribution of $\bar{x}$

Show that sample mean = population mean
 calculate sample variance.

2, 4, 6, 8 condition $\Rightarrow$ (with replacement)

$N = 4$

$n = r = 2$

$${}^n C_r = (n)^r \Rightarrow (4)^2 \Rightarrow 16$$

Sample #	sample	$\bar{x}$
1	(2, 2)	$\frac{2+2}{2} = 2$
2	(2, 4)	3
3	(2, 6)	4
4	(2, 8)	5
5	(4, 4)	4
6	(4, 2)	3
7	(4, 6)	5
8	(4, 8)	6

9	(6,6)	6
10	(6,2)	4
11	(6,4)	5
12	(6,8)	7
13	(8,8)	8
14	(8,2)	5
15	(8,4)	6
16	(8,6)	7
		<u>125</u>

$\bar{x}$	f	$x_i f_i$
2		
3		
4		
5		
6		
7		
8		
		<u>$\Sigma x_i f_i$</u>

$$\bar{x} = \frac{\Sigma x_i f_i}{\Sigma f_i} \Rightarrow$$

$$\mu = \frac{2+4+6+8}{4} \Rightarrow 5$$

$$V(\bar{x}) = \frac{s^2}{n}$$

$$\therefore s^2 = \frac{\Sigma x^2}{N} - \left(\frac{\Sigma x}{N} \right)^2$$

$$= \frac{120}{4} - \left(\frac{20}{4} \right)^2$$

$$\boxed{s^2 = 5}$$

$$\Rightarrow \frac{5}{2}$$

$$\Rightarrow 2.5$$

for s^2 formula

x	x^2
2	4
4	16
6	36
8	64
<u>120</u>	

If sample size is greater than population size
so n will be smaller value

Q) All possible sample of size = 3 from the population
~~2, 11~~ 2, 11 with replacement.

2, 11

$$N = 2$$

$$n = 2$$

$$r = 3$$

$$nCr = (n)^r = (2)^3 = 8$$

Sample #

			x	f	$xi \cdot f$
1	(2, 2, 2)	2			
2	(2, 2, 11)	5	2	1	2
3	(2, 11, 11)	8	5	3	15
4	(2, 11, 2)	5	8	3	24
5	(11, 11, 11)	11	11	1	11
6	(11, 11, 2)	8		8	52
7	(11, 2, 2)	5			
8	(11, 2, 11)	8			

$$\bar{x} = \frac{\sum xi \cdot f}{n}$$

~~na~~

Testing may be defined as a process in which sample data are used to decide whether the statement is accept or reject

Testing may consist following steps.

1. Null hypothesis H_0 :
2. Alternative hypothesis H_1 or H_a :
3. Level of significance $LoS = \alpha$
4. Test Statistics $\Rightarrow$ formula used for z test $Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$
5. Critical region (CR)
6. Conclusion

(z test, t test, chisquare test (χ^2))

if $(n = \infty)$ n is large then we will use z test
 $(n \leq 10)$ less than equal to 10 then t test will be use.
 qualitative data then chisquare will use.

z test is of two types

① onetail

critical region will be

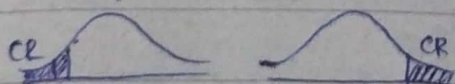
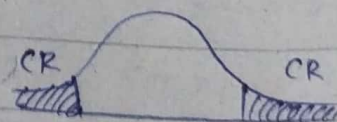


Table will be given

② twotail



	$LoS = \alpha$	0.10	0.05	0.01
onetail	$Z_{\alpha} = \pm 1.28$	$Z_{\alpha} = \pm 1.65$	$Z_{\alpha} = \pm 2.33$	
two tail	$Z_{\alpha/2} = \pm 1.65$	$Z_{\alpha/2} = \pm 1.96$	$Z_{\alpha/2} = \pm 2.58$	

Q) A machine is set to filled CR box with mean weight of 500gms. The standard deviation is 25gm and random sample of 100 box is taken from the process. The mean weight of CR box is 504gm is there any reason to believe that the machine is overfilling the boxes used $\alpha = 0.05$

$$n = 100$$

$$s = 25$$

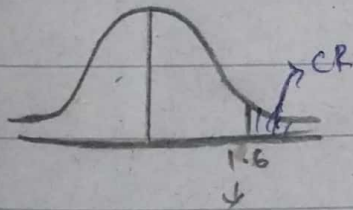
$$\bar{x} = 504$$

$$H_0: \mu \leq 500$$

$$H_a: \mu \geq 500$$

$$\alpha = 0.05$$

$$T.S = Z = \frac{\bar{x} - \mu}{s/\sqrt{n}} \Rightarrow \frac{504 - 500}{25/\sqrt{100}} \Rightarrow 1.6$$



calculated values

$$Z_{cal} = 1.6$$

$$Z_{Tab} = 1.65$$

$$Z_{1.65} > Z_{cal 1.6}$$

calculated value 1.6 hai or table say 0.05 say 1.65 aisi hai to 1.6 ko likhengey bec 1.65 to CR kay andar jayahi hai.

agar cal value badi ho cal value say so we accept H_0 and rej H_a

Statement :

Since Z calculated less than less tabulated so we accept H_0 and reject H_1 on the other word we can say that H_0 is fail to pass

~~Q1~~ (Q) A random Sample of size 1089 from a normal population with standard deviation 2.6. If the mean of the sample is 3.60. Test the hypothesis that $\mu = 3.90$ used $\alpha = 0.10$

$$\textcircled{1} H_0 : \mu = 3.90$$

$$H_a : \mu \neq 3.90$$

$$\text{los} = \alpha = 0.10$$

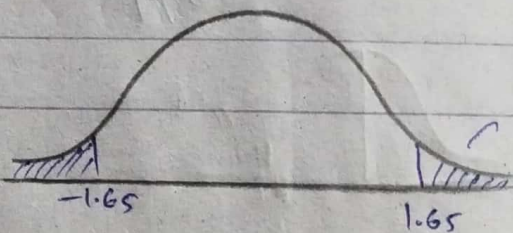
using formula of Test of statistics:

$$Z = \frac{\bar{X}_a - \mu}{s/\sqrt{n}}$$

$$\Rightarrow \frac{3.6 - 3.90}{2.6/\sqrt{1089}}$$

$$\left[\begin{array}{l} Z_{\text{calculated}} \\ \Rightarrow -3.81 \end{array} \right]$$

$$\boxed{Z_{\text{Tab}} = -1.65}$$



Since calculated value of z fall in the critical region therefore we reject null hypothesis and accept alternative hypothesis

T-TEST :

$$n \leq 30$$

$$\text{Degree of freedom} = D.F = \gamma = n - 1$$

(Q) A random sample of size 20 from a normal population as mean 16.4 and standard deviation is 2.255 does this suggest at the $\alpha = 0.05$ level of significance that the population mean is greater than 15

sigma (σ) in z test
s is used in t-test

$$H_0: \mu \leq 15$$

$$H_a: \mu > 15$$

$$\alpha = 0.05$$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \Rightarrow \frac{16.4 - 15}{2.255/\sqrt{20}}$$

$$t = 2.775 \rightarrow t_{\text{calculated}}$$

for Tabulated value we will first find D.F

$$D.F = n - 1$$

$$\Rightarrow 20 - 1 \Rightarrow 19$$

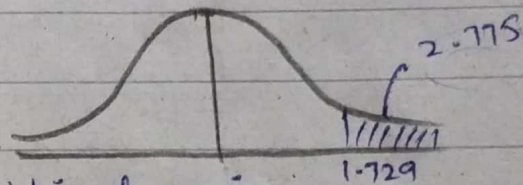
Now from the t table in α column we will see the value in 0.05 and in γ 19
so

$$t_{\text{tab}}(19, 0.05) = 1.729$$

Since the calculated value of

T distribution fall in critical region

so we reject H_0 and accept alternative H_a and
we conclude that the population mean is greater
than 15



Q) A random sample of 16 observation from a normal population with $SD = 24$ and mean is 110 test the hypothesis that $H_0 = 100$ and alternative $\mu > 100$
 $\alpha = 0.05$

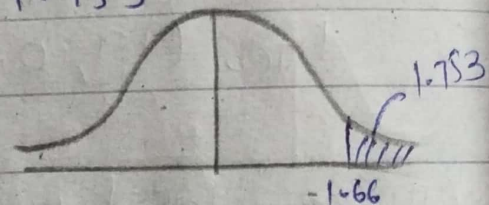
$$H_0: \mu = 100, \quad H_a: \mu > 100 \quad \alpha = 0.05$$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \Rightarrow \frac{110 - 100}{24/\sqrt{16}}$$

$$t_{cal} = 1.66$$

Now, $D.F = \gamma = n - 1$
 $= 16 - 1 \Rightarrow 15$

$$t_{tab} = (15, 0.05) \Rightarrow 1.753$$



Q) A random sample of size 9 of peanut jar was taken from a lot each jar has been labelled 10.5gm net weight. The individual weight of the jar are 10.2, 94.09, 102.01, 98.01, 106.09, 90.25, 92.16, 106.09, 108.16, 9.7, 10.1, 9.9, 10.3, 9.5, 9.6, 10.3, 10.4. Test the hypothesis whether these values are considered with the population of weight of mean 10.5 assume that the weight of the jar is normally distributed using $\alpha = 0.05$

in two tail 2α will be $2\alpha/2$
So 0.05 will be 0.025

$$n=9$$

$$H_0: \mu = 10.5$$

$$H_a: \mu \neq 10.5 \quad \alpha = 0.05$$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \Rightarrow$$

First we will find SD

$$s = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

$\Rightarrow$

$$\Rightarrow \frac{10 - 10.5}{0.335/\sqrt{9}}$$

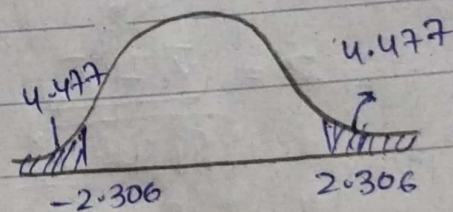
$$T_{cal} = -4.477 \Rightarrow |-4.477| \Rightarrow \boxed{4.477}$$

$$D.F = n - 1 = 8$$

$$t_{tab} = (8, 0.05) = 1.860$$

$$t_{tab} = (8, 0.025) = \boxed{2.306}$$

beg two tail so $\alpha/2$



Conclusion :

We reject null hypothesis

to and accept H_a because
calculated T is greater

Testing almost two means

$$H_0 : \mu_1 = \mu_2 \quad \boxed{\text{OR}} \quad \mu_1 - \mu_2 = 0$$

$$H_a : \mu_1 \neq \mu_2 \quad \boxed{\text{OR}} \quad \mu_1 - \mu_2 \neq 0$$

$$\alpha = 0.05 \quad / \quad 0.10 \quad / \quad 0.01$$

$$z = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

z-testing is used bcz n value is greater than 30
Q) A random sample of size $n_1 = 50$ taken from normal population with $SD = s_1 = 7.35$ and mean $= \bar{x}_1 = 181$
a second random sample of size $n_2 = 72$ with $SD = s_2 = 4.81$ and $\bar{x}_2 = 176$. Test the hypothesis at the level of 0.05 significant that $\alpha = 0.05$

$$H_0 : \mu_1 = \mu_2 \quad , \quad H_a : \mu_1 \neq \mu_2$$

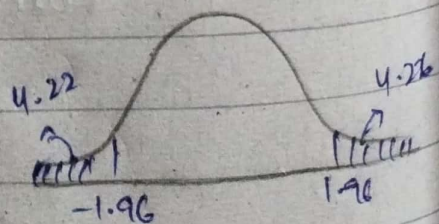
$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \Rightarrow \frac{181 - 176}{\sqrt{\frac{(7.35)^2}{50} + \frac{(4.81)^2}{72}}}$$

$$z_{cal} = 4.22$$

two tail testing so α will be $\alpha/2$ so

$$z_{tab} = 1.96$$

We reject null hypothesis (H_0)
because it falls in critical region
and accept alternative hypothesis (H_a)



t -Test will be used bcz n value is less than 30

Q) From two random samples of size $n_1 = 11$ and size $n_2 = 14$ from two independent normal population with $\bar{x}_1 = 85$ and $\bar{x}_2 = 70$ and $SD = s_1 = 6.1$ and $s_2 = 5.3$. Test the hypothesis

$$H_0: \mu_1 = \mu_2, \quad H_a: \mu_1 \neq \mu_2 \quad \alpha = \frac{0.005}{2}$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{85 - 70}{\sqrt{\frac{(6.1)^2}{11} + \frac{(5.3)^2}{14}}}$$

$$SP = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$
$$t = \frac{\bar{x}_1 - \bar{x}_2}{SP \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

if only one value is given then -1

NOW

$$SP = \sqrt{\frac{(11-1)(6.1)^2 + (14-1)(5.3)^2}{11+14-2}}$$

$$\Rightarrow 5.66$$

$$t_{cal} = \frac{85 - 70}{5.66 \sqrt{\frac{1}{11} + \frac{1}{14}}} \Rightarrow 6.57$$

$$D.F = n_1 + n_2 - 2$$
$$\Rightarrow 11 + 14 - 2 \Rightarrow 23$$

$n \rightarrow \alpha$ large

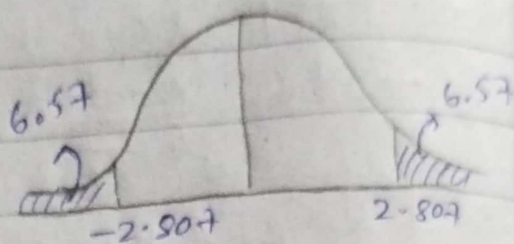
$p \rightarrow$ small

$b \sim p$ (binomial approximate to poisson)

$np = \lambda$ poisson

binomial distribution
main np
binomial ka
mean hai
 $\bar{x} = np$

$$T_{tab}(0.005, 23) \Rightarrow 2.807$$



Q) Fit a binomial distribution to the following dataset

Suppose that a coin is biased and tossed 5 times

Given data: (show that the data is a good fit in binomial)

H	f	xf
0	12	
1	56	
2	74	
3	39	
4	18	
5	1	
	200	

$\Rightarrow \frac{0 \times 12 + 1 \times 56 + 2 \times 74 + 3 \times 39 + 4 \times 18 + 5 \times 1}{200}$

$\bar{x} = 1.99 \Rightarrow np$

$p = \frac{1.99}{5}$

$p = 0.39$

$q = 1 - 0.39$

$q = 0.61$

notation of binomial $b \sim (x: np)$
 $b \sim (5: 50.38)$

$200 \cong 199$ if ans is like this then ans will be good fit
 if $200 \cong 196$ then will be bad fit

# x	$bn(x:np) = \binom{n}{x} p^x q^{n-x}$	$E(\bar{x})$ $\sum fx p(x)$
0	$\binom{5}{0} (0.39)^0 (0.61)^5 = 0.08$	$200(0.08) = 16$
1	$\binom{5}{1} (0.39)^1 (0.61)^4 = 0.26$	
2	$\binom{5}{2} (0.39)^2 (0.61)^3 =$	
3	$\binom{5}{3} (0.39)^3 (0.61)^2 =$	
4	$\binom{5}{4} (0.39)^4 (0.61)^1 =$	
5	$\binom{5}{5} (0.39)^5 (0.61)^0 =$	